

Quick Rad AI - On-Prem DICOM ANONYMISER

Deployment & Security Guide for Hospital IT - GenZ AI Labs

This device is LOCAL-ONLY. It receives DICOM studies from your scanner, removes patient identifiers ON THIS PC, and writes anonymised studies to a folder that a technician uploads manually. It makes NO outbound network connection - no PHI is transmitted by this software. The claims below are backed by the verification results (last section), not asserted.

1. Data flow

Scanner --C-STORE--> [this PC : 11120, AE QR_ANONYMISER] --> staging (temporary, encrypted volume) --> anonymise (PS3.15 + UID remap) --> qr_data\output\<new_uid>\ + manifest.txt. The staging original is deleted after anonymisation. A technician uploads the output manually. No outbound connection anywhere.

2. What is removed and what is kept

Action	Tags
REMOVED	Patient name, MRN, date of birth, address, other IDs, patient comments; referring/performing physician, operator, institution, station name, device serial; free-text.
KEPT	AGE and SEX; accession number (technician link); ALL clinical/technical tags - modality, series/protocol, TR, TE, IR, FOV, kV, mAs, image number, geometry, dates & times.
KEPT (vendor)	Diffusion b-value private tags: Siemens 0019,100C; GE 0043,1039; Philips 2001,1003. All other private tags removed.
REMAPPED	Every Study/Series/SOP/FrameOfReference UID and referenced UID is replaced. Original UIDs never appear in output tags or filenames.
SKIPPED	SecondaryCapture screenshots are not uploaded.

3. Network behaviour

Direction	Host	Port	Purpose
Inbound (LAN only)	this PC	TCP 11120	receive studies (AE QR_ANONYMISER)
Outbound	NONE	-	this software never connects out

The installer opens only an INBOUND firewall port. There is no outbound rule and no cloud endpoint.

4. Package contents

File	Purpose
qr_anonymiser.py	the local receiver + pipeline (listens only, no egress)
anonymise.py	PS3.15 profile + consistent UID remap + b-value whitelist
guard.py	SecondaryCapture detection + platform ingest safety check
anonymiser_config.json	ports, AE title, whitelist, folders (no cloud fields)
install.bat / run.bat	install as auto-start job (inbound firewall only) / console test
requirements.txt / README.txt	dependencies (pynetdicom, pydicom) / instructions

5. Verification (evidence, not claims)

Results below were produced by running the software against a synthetic neuro study (no patient data).
Re-runnable on request.

Invariant / check	Result
I1 - No outbound network calls (static scan + runtime with connect() disabled)	PASS
I2 - Original Study/Series/SOP UIDs never leave output (tags or filenames)	PASS
I3 - Vendor diffusion b-value private tags survive	PASS
Seeded patient identifiers absent from output	PASS
Age & sex preserved	PASS
Accession number preserved (technician link)	PASS
A/B clinical: pixels + diffusion/geometry byte-identical before vs after	PASS
Platform guard accepts every anonymised instance	PASS

Static egress scan hits: 0. b-values before ['0', '1000'] == after ['0', '1000']. Original UIDs found in output: 0.

OVERALL: ALL CHECKS PASS

6. Install / manual upload / remove

- Install Python 3.12 (Add to PATH). Unzip. Right-click install.bat -> Run as administrator.
- Scanner destination: Called AE QR_ANONYMISER, this PC's LAN IP, port 11120. DICOM Echo, send.
- Anonymised studies appear in qr_data\output\<new_uid>\ with manifest.txt (new-UID, accession, age, sex, series). A technician uploads that folder.
- Put qr_data\ on a BitLocker (encrypted) volume so the brief staging copy is encrypted.
- Remove: schtasks /delete /tn "QuickRadAnonymiser" /f and delete the inbound firewall rule ("QR Anonymiser DICOM").

GenZ AI Labs - Quick Rad AI. Anonymisation implements the DICOM PS3.15 Basic Application Level Confidentiality Profile on pydicom. This is the on-prem device; cloud-side retention/hosting/access are documented separately.